VIERA KOVACOVA PhD



DATA SCIENCE & BIOINFORMATICS LEADER

Cologne, Germany

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15+ years delivering computational solutions across biomedical research and molecular diagnostics. Co-PI leading a central bioinformatics function for 15+ research groups across theoretical physics, experimental biology and translational medicine. Combines predictive modelling, rigorous validation and reproducible platform development with scientific consulting, mentoring and industry R&D leadership.

PROFESSIONAL EXPERIENCE

Co Principal Investigator

Jan 2026–present | CRC1310, University of Cologne

- Lead the central bioinformatics function of a DFG-funded research consortium; translate scientific priorities into computational methods, workflows and data standards.
- Advise 15+ groups on experimental design, statistical modelling, data quality and reproducible analysis; mentor researchers and coordinate delivery.

Lead Bioinformatician

2018–2020; 2022–2025 | University of Cologne

- Built RNA-seq, WGS and phylogenomics workflows on HPC; implemented consortium metadata standards and research data management.
- Evolutionary modelling of immune receptor modules predicting novel protein interactions.

Head of Bioinformatics in R&D

2020–2021 | MultiplexDX, Bratislava

Continued as short-term contractor, 2022–2023.

- Led computational work for breast cancer diagnostics and SARS-CoV-2 surveillance; designed variant-tracking RT-qPCR assays reported in Scientific Reports (2021).
- Secured PRACE SHAPE HPC funding and delivered analyses under pandemic timelines across R&D and public-health requirements.

Bioinformatician

2014–2018 | CECAD, University of Cologne

Genomics, transcriptomics, phylogenetic modelling and protein interaction prediction.

EDUCATION

PhD General and Molecular Genetics

Masaryk University & Czech Academy of Sciences, Brno | 2010–2017

MSc and BSc Biotechnology

Brno University of Technology | 2005–2010

LANGUAGES

Slovak native · English fluent
German and Hungarian intermediate

SELECTED DATA SCIENCE DELIVERIES

Long COVID predictive diagnostics

Built interpretable Firth logistic classifiers with bootstrap validation, calibration, noise stress tests and independent-cohort evaluation. Evaluated two- and three-protein panels; external AUC 0.901 and 0.931, respectively.

Microbiome machine learning

Applied random forests and feature selection to body-site classification; assessed OOB performance, ROC curves, mixed-effects sensitivity and leave-one-horse-out validation.

RNA degradation and model reliability

Designed ground-truth simulations, sequencing-derived features and predictive GLMs. Compared nonlinear response models and tested unseen severity regimes, identifying extrapolation failures and model-fitting artifacts.

PhAMiG analytical platform

Sole developer of a Shiny/DuckDB growth-data platform: 12,965 datasets across 30 species. Built import adapters, metadata curation, controlled release and benchmarking of growth-analysis tools under consistent preprocessing.

METHODS AND TECHNOLOGIES

Modelling

Penalized logistic regression, LASSO, random forests, GLMs and mixed models, nonlinear curve fitting, negative-binomial RNA-seq models, phylogenetics and network analysis.

Validation and quality

Bootstrap and held-out validation, calibration, uncertainty, confounding and bias assessment, simulation benchmarks and sensitivity analyses.

Engineering

R, Python, Bash, Git, Nextflow, Linux/HPC, Shiny, DuckDB; prior AWS experience.

LEADERSHIP AND COMMUNICATION

Scientific consulting and cross-functional R&D delivery; grant writing and mentoring. Invited lecturer, CodeBridge Biomed Data School (2025–2026); Aj Ty v IT tutor (2020–2023).